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The Role of Psychosocial Factors in Returning to Physical Activity and Sport Among Female Football Players After Anterior Cruciate Ligament Injury: A Scoping Review

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Abstract

Background Anterior cruciate ligament (ACL) injuries are an increasingly common knee injury in sport requiring quick direction changes, such as the football codes. Return to physical activity and sport following an ACL injury predominantly focuses on achieving objective milestones to resolve physical impairments, with sex differences reported. However, irrespective of management approach following an ACL injury, many athletes do not return to the same level of sport, if at all, warranting the need to consider psychological and social factors.

Aim The aim of this scoping review was to investigate the role of psychosocial factors in females participating in any football code after an ACL injury, and the impact on return to physical activity and sport.

Methods A systematic search of six electronic databases (SCOPUS, CINAHL, MEDLINE, SPORTDiscus, PsycInfo, and Embase) was conducted on 1 May 2025. Two reviewers independently screened articles against the eligibility criteria to include original research articles that explored the impact of psychosocial factors on return to sport (RTS) or physical activity in females playing any football code with an ACL injury managed surgically or non-surgically. Studies were eligible where participants had a mean age of 18 years or older. Data extraction and narrative synthesis of included studies was undertaken and themes identified to address the research aim.

Results Full-text articles ($n=74$) were assessed for eligibility following screening of titles and abstracts ($n=582$); seven eligible studies were included. Across the studies, psychosocial barriers and facilitators were perceived to influence RTS at either preinjury or lower competition levels, as well as physical activity, in female footballers. Uncertainty, fear, lack of confidence, and poor social support were identified as perceived barriers to RTS contributing to return at a lower level of competition and the alteration, limitation, or avoidance of physical activity among female footballers. Whilst confidence, motivation, strong social support and athletic identity were identified as perceived facilitators of RTS, helping to build readiness and supporting female footballers to return at their preinjury level or a lower level of competition.

Conclusion Psychosocial factors appear to play an important role in RTS and physical activity following ACL injury in female football players, acting as both barriers and facilitators to recovery. Further high-quality research focussed specifically on female footballers is needed to better understand these relationships and inform targeted screening and interventions.

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Key Points

- Psychosocial factors have a perceived impact on female football players returning to sport and/or physical activity after ACL injury, acting as barriers and facilitators.
- Uncertainty, fear, lack of confidence, and poor social support were identified as perceived barriers to return to sport, contributing to return at a lower level of competition and the alteration, limitation, or avoidance of physical activity among female footballers.
- Confidence, motivation, strong social support and athletic identity were identified as perceived facilitators of return to sport, supporting female footballers to return at their preinjury level or a lower level of competition.
- Incorporating psychosocial screening into rehabilitation may help support return to sport outcomes in female footballers following ACL injury.

Keywords ACL injury, Women's football, Soccer, Return to sport, Fear of reinjury, Confidence

Background

Anterior cruciate ligament (ACL) injuries are an increasingly common knee injury in sport requiring quick direction changes, such as in the various sport known as football; these include soccer (association football) [1], the rugby codes (union and league) [2], the national codes of Australian football [3, 4], American football [5], and Gaelic football [6]. In both males and females, ACL injury rates have doubled over the last two decades despite increased research and injury prevention programmes [7], with Australia having the highest and fastest growing rate of ACL reconstruction (ACLR) surgery performed annually [8].

Female participation in sport has increased more than ten times over the past four decades [9]. This cohort has had a corresponding rise in ACL injuries and a different risk profile to males [10]. ACL injury risk in females is 2–8 times greater than that of men in team sport [7, 10]. For example, the incidence rate of ACL injuries in female soccer players is 2.5 times higher than in male athletes, with many of these injuries occurring in females under 30 years of age [10, 11]. Similarly, female Australian footballers participating in the Australian Football League Women's (AFLW) and non-elite competitions, have a relative risk of ACL injury (adjusted for season length and player exposure hours) that is 6.6 times higher than that of male Australian footballers [4]. Not only do female athletes have a higher injury risk than male athletes, but emerging research has found that they also report reduced activity and knee-related outcomes after ACL injury compared to male athletes [12]. Female footballers navigate a sociocultural environment marked by structural, financial, and medical disadvantage relative to their male counterparts [12]. Compared to male footballers, elite female footballers AFLW have lower pay and reduced access to allied health support and facilities, while balancing training with employment and family commitments [13]. Together, the heightened risk of ACL injury and the unique sociocultural challenges experienced by female footballers highlight the need for research examining factors that influence recovery following ACL injury.

Ensuring female athletes return to full physical activity and return to sport (RTS) is a key consideration after ACL injury or reconstruction [12, 14]. Irrespective of management approach following an ACL injury, many athletes do not return to the same level of sport, if at all [15]. In a study by Ezzat et al. [14], only 37.3% of female athletes returned to the same level of sport played before their ACL injury and only 56% reported being satisfied with physical activity levels. Recovery from an ACL injury can include surgical or non-surgical management and predominantly focuses on achieving objective milestones to resolve physical impairments in preparation for return to physical activity and sport [15], not often delving into psychological and social considerations. The Biopsychosocial (BPS) Model of Athletic Injury Rehabilitation is a theoretical framework for understanding key components influencing sport injury rehabilitation and outcomes. Originally proposed by Brewer in 2001 [16], the model consists of seven components that conceptualise rehabilitation as the dynamic interaction of biological, psychological, and social factors [17]. When an individual experiences a sport injury, the injury characteristics (such as injury type, severity, cause, and location) and sociodemographic factors (such as age, gender, and socioeconomic status) shape various biological, psychological and contextual factors that influence the rehabilitation process [17]. Biological factors such as tissue repair processes, sleep, and immune function contribute to physical recovery and interact with psychological and social factors throughout the rehabilitation process [17, 18]. Social factors are present within situational and environmental contexts, and include social network, life stress, situational characteristics, and rehabilitation

environments [17, 19]. Psychological factors refer to cognitive processes, such as individual personality traits, as well as behavioural and affective responses, such as mood [17, 19]. Within the BPS Model of Athletic Injury Rehabilitation, psychological factors have reciprocal relationships with biological and social factors throughout the rehabilitation process [17]. These interactions can modify behaviours [20]. For example, Williams et al. [20] found that optimism in athletes recovering from ACL injury drove rehabilitation adherence and improved knee-related outcomes, such as strength. Additionally, limited social support may contribute to emotional distress and fear of reinjury, which can prolong rehabilitation engagement and delay RTS [21]. Through these interactions, biological, psychological, and social factors influence intermediate biopsychological outcomes such as strength and pain, which in turn impact broader sport injury rehabilitation outcomes, including readiness to RTS [17]. Therefore, psychosocial factors should be considered alongside physical objective milestones after ACL injury [15].

Previous reviews have examined the role of psychosocial factors during recovery following traumatic time-loss knee injuries [15], and specifically, ACL injury [22] in male and female athletes. Findings revealed psychological factors, such as fear, coping, and self-efficacy, were identified as barriers or facilitators to recovery from a knee injury [15, 22]. In a scoping review by Truong et al. [15] examining psychological, social, and contextual factors in mixed sex athletes with traumatic knee injury, few studies explored social and contextual factors (39% and 21%, respectively, of 77 included studies) indicating a limited understanding of the role of social and contextual factors in recovery and RTS. Additionally, a recent meta-analysis by Bruder et al. [12] revealed sex differences exist in self-reported activity and knee-related outcomes following an ACL injury; females/women/girls had inferior outcomes to males/men/boys. Therefore, investigating the role of psychosocial factors following ACL injury specific to females is crucial to understand differences in ACL injury recovery experiences. Psychological factors together with the unique sociocultural challenges female footballers face throughout their careers and after ACL injury, is an underexplored knowledge gap that needs to be adequately addressed to support females returning and staying in sport. As such, further research is needed to explore a broad range of non-physical influences that inhibit, impact, or limit progression back to full physical activity and sport in female athletes participating in football codes. Therefore, the aim of this scoping review was to investigate the role of psychosocial factors in return to sport and physical activity in female athletes participating in any football code after an ACL injury.

Methods

A scoping review design is a form of evidence synthesis that identifies, maps, and summarises the breadth of literature on a particular topic [23]. This method was selected over a systematic review because the aim was to comprehensively map the evidence relating to psychosocial factors influencing ACL injury recovery and identify knowledge gaps to inform future research and better support female footballers. This scoping review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) extension for scoping reviews (PRISMA-ScR) [24]. The PRISMA-ScR checklist is included in Supplementary Material. An exploratory search for similar systematic/scoping reviews and published protocols was conducted before commencement to ensure no overlapping evidence synthesis. This scoping review was registered with Open Science Framework (OSF) on 14 April 2025 (https://doi.org/10.17605/OSF.IO/PEFX6).

Eligibility Criteria

Eligibility criteria for this scoping review were established using the Population, Concept, Context (PCC) framework [25]. The inclusion and exclusion criteria for this scoping review are outlined in Table 1. This review focussed on studies that included female athletes aged between 18 and 65 years, playing one of the football codes (soccer, rugby, Australian football, American football, and Gaelic football) at any level of participation, with an ACL injury managed surgically or non-surgically and investigated the impact of psychosocial factors on physical activity, exercise, or RTS. Table 2 includes definitions of these key terms used in this scoping review. The age criterion ($\geq$ 18 years) was revised post-hoc to include studies with some younger participants where the sample mean age remained 18 years or older. The authors recognise the differences in sex and gender [26]; throughout this review, we refer to female participants for consistency with all included studies.

Children and adolescents were excluded from this review. The review was restricted to skeletally mature female football players because biological maturation is an important determinant of ACL injury risk, rehabilitation, and RTS outcomes. Skeletally immature athletes demonstrate substantial inter-individual variability in growth, maturation, and pubertal development [27, 28], while hormonal and musculoskeletal changes occurring during adolescence may influence both injury mechanisms and recovery trajectories [29–31]. In addition, psychosocial factors associated with RTS, including autonomy in decision-making, athletic identity, family involvement, and educational commitments, differ considerably between adolescent and adult populations [32,

Table 1 Inclusion and exclusion criteria for studies investigating psychosocial factors in female footballers after ACL injury

Inclusion criteria	Exclusion criteria
Females/women Athletes where the mean age of study participants was 18 years or older	Female players subset data not reported separately Youth athletes (< 18 years old) only or where the mean age of study participants fell below 18 years of age or when adult subset of data not reported separately Non-human studies
Includes at least one of the football codes: soccer, rugby, Australian football, American football, and Gaelic football Any level of participation	When football subset of data not reported separately
ACL injury managed surgically or non-surgically, including combination knee injuries	Non-ACL knee injury; knee injury diagnosis unclear where multiple knee injuries are investigated; ACL injury subset of data not reported separately
Investigates impact of psychosocial factors on return to physical activity, exercise, or RTS	
Primary research including quantitative, qualitative and mixed methods, published in English	Not primary study design; protocols, editorials, conference abstracts and grey literature

RTS: return to sport; ACL: anterior cruciate ligament

Table 2 Working definitions for key terms used in this scoping review

Term	Definition
Psychological factors	Considers meanings and perceptions related to cognitive, behavioural and affective responses influencing an athlete's mental state [15]
Social factors	Examines how societal structures, networks, support systems and exchanges, processes, and expectations impact individuals' experience [15]
Physical activity	Moderate-to-vigorous physical activity (MVPA) refers to physical activity that is performed at three times the intensity of rest, based on the individual's personal capacity [34]. The World Health Organisation (WHO) recommendations include an average of 60 min of MVPA daily for individuals 5–17 years of age; and 150–300 min of moderate intensity or 75–150 min of vigorous intensity, or an equivalent combination of MVPA for individuals 18–64 years of age
Return to sport	Return to sport refers to returning to competitive football at preinjury level, including Australian football, soccer, rugby, American football, or Gaelic football, after ACL injury managed surgically or nonsurgical. This encompasses a graduated rehabilitation program aimed at restoring knee function, strength, plyometrics, and neuromuscular control [35], return to participation and training, before returning to football [36]

ACL: anterior cruciate ligament

33]. Given these differences were considered likely to influence RTS outcomes, studies involving paediatric or adolescent athletes where the mean age was less than 18 years old were excluded to reduce heterogeneity and maintain population-specific relevance.

Search Strategy

We searched six electronic databases on May 1, 2025: SCOPUS (Elsevier), CINAHL (EBSCO), MEDLINE (EBSCO), SPORTDiscus (EBSCO), PsycInfo, and Embase (Elsevier) from inception. In addition to database searches, reference lists from included articles were screened. The final search strategy was developed in consultation with an experienced Health Science and Medicine librarian from Bond University. The search strategy included MeSH terms and keywords relating to the research question using the PCC framework for scoping reviews [25]. Search terms associated with female athletes (18 years and older) playing any football code at any level of competition (P), sustaining an ACL injury requiring surgical or non-surgical management (C), and explored psychosocial factors and physical activity or RTS (C) were included in the search strategy (Supplementary Material). A comprehensive list of psychosocial factors was adapted from a 2018 study by Truong et al. [15] to form the search strategy. The final search strategy for Medline database is available in the Supplementary Material. This final search strategy was then modified to support the syntax of remaining databases searched (Supplementary File 3).

Study Selection

All identified articles from database searches were collated and uploaded into EndNote 21 (Clarivate Analytics, PA, USA). Articles were then imported into Covidence (Veritas Health Innovation, Melbourne, Australia), and duplicates removed. Title and abstract screening were completed in Covidence by two independent reviewers (SM and ED) against the eligibility criteria for the review. Relevant articles were retrieved as full texts and assessed

in detail against the eligibility criteria by two independent reviewers (SM and ED). Reasons for exclusion at the full-text stage were recorded and reported in Covidence. Reference lists of included studies were screened for additional studies. Any disagreements were resolved through discussion or with an additional reviewer (JF).

Data Extraction and Analysis

Data was extracted from eligible studies using a predetermined data extraction table into a custom Microsoft Excel (Microsoft Corporation, Redmond, WA, USA) spreadsheet by an independent reviewer (SM). The spreadsheet was tabulated using the following headings: author, year, country, study design, population, age at injury, age at surgery or time since surgery, sport and competition level, ACL injury management, study method, psychosocial factor(s) and physical activity/return to sport outcomes investigated, and key findings relating to the research question of this review. The data extraction was pilot tested on three studies to identify potential inconsistency and ensure reliability. The final data extraction table was reviewed by an additional reviewer (JF) for accuracy and consistency. Any disagreements between the reviewers were resolved through discussion with a third reviewer (ED). Authors of papers were contacted to request missing data. If demographic data was available for each participant, the first author (SM) calculated mean and standard deviation (SD). A critical appraisal of included studies is not required for scoping reviews [23], therefore was not conducted.

To summarise the findings of the review, a narrative synthesis was conducted. Psychosocial factors were first identified independently by two authors (SM and JF) through structured review of the data extraction table, with recurring factors highlighted across one or more included studies. Through author discussion, themes highlighted in data relating to female footballers were identified. Findings were synthesised alongside RTS and physical activity outcomes and/or participant quotations to classify psychosocial factors as a barrier and/or facilitator. Sample quotations from the data extraction table are provided to demonstrate the participant wording and expressed views underpinning the identified factors [37].

Results

Study Selection and Characteristics

The search strategy produced 1,170 articles with one additional article identified following reference list screening. Seven studies were included in the review for analysis (Fig. 1). Summary of study characteristics are presented in Table 3. Of the included articles, five (71%) implemented qualitative methodology using focus groups, semi-structured interviews, or open-ended surveys to investigate the impact of psychosocial factors on return to physical activity or RTS. Two (29%) studies utilised quantitative study design using questionnaires. To measure physical activity or indicators for RTS, three studies [38–40] used the Anterior Cruciate Ligament Return to Sport after Injury scale (ACL-RSI) and the Tegner Activity Score (TAS), one study used only the TAS [41], and three studies used questions related to physical activity and/or sport [42–44]. The ACL-RSI is a highly consistent and reproducible scale that is used to assess psychological readiness for RTS following ACL injury, measuring factors such as confidence, emotions, and risk appraisal, with a high score indicating more psychological readiness [45]. The TAS grades activities on a scale of 1–10, with 10 being participating in football on a national or international level [39].

A total of 403 female participants ranging from age 16–56 were included; all studies included athletes who played soccer, with one study including one participant who played rugby union [40] and another study including one participant who played touch football [42]. All participants played football at the time of ACL injury. Of the 403 included female football players, 222 (55.1%) RTS while 181 (44.9%) did not RTS after ACL injury. Among those who RTS, all resumed participation in football except one player who switched to a different sport [42]. Of those that did RTS, competition level ranged from elite, international athletes to recreational, local female footballers [38–40, 42, 43]. Among female footballers who returned to preinjury level of competition, two competed at the elite level [40], one at the state level [43], one at the school/university level [43], and three at the local level [43]. Among those who returned to a lower level of football competition post ACL injury, one player transitioned from national to regional level competition [43], three went from state to local or regional competition levels [43] and the postinjury competition level was not specified for one player [42]. The remaining 210 players who returned to football had insufficient information available to determine their postinjury competition status relative to their preinjury participation level [38, 39]. Of the female footballers who did not return to sport, 10 players (5.5%) returned to physical activity such as running, walking, or cycling; however, they were not able to return to self-reported desired physical activity levels [41, 44]. Physical activity was measured by self-reported TAS [38, 39, 41], participant quotations [42, 44] or questions such as "are you satisfied with current activity level" [38, 39]. Time frames post ACL injury or reconstruction ranged from 1.5 to 24 years at time of study. Six (86%) studies included both male and female athletes, with female participant data reported separately extracted for this review [38, 40–44]. Of these mixed sex-studies,

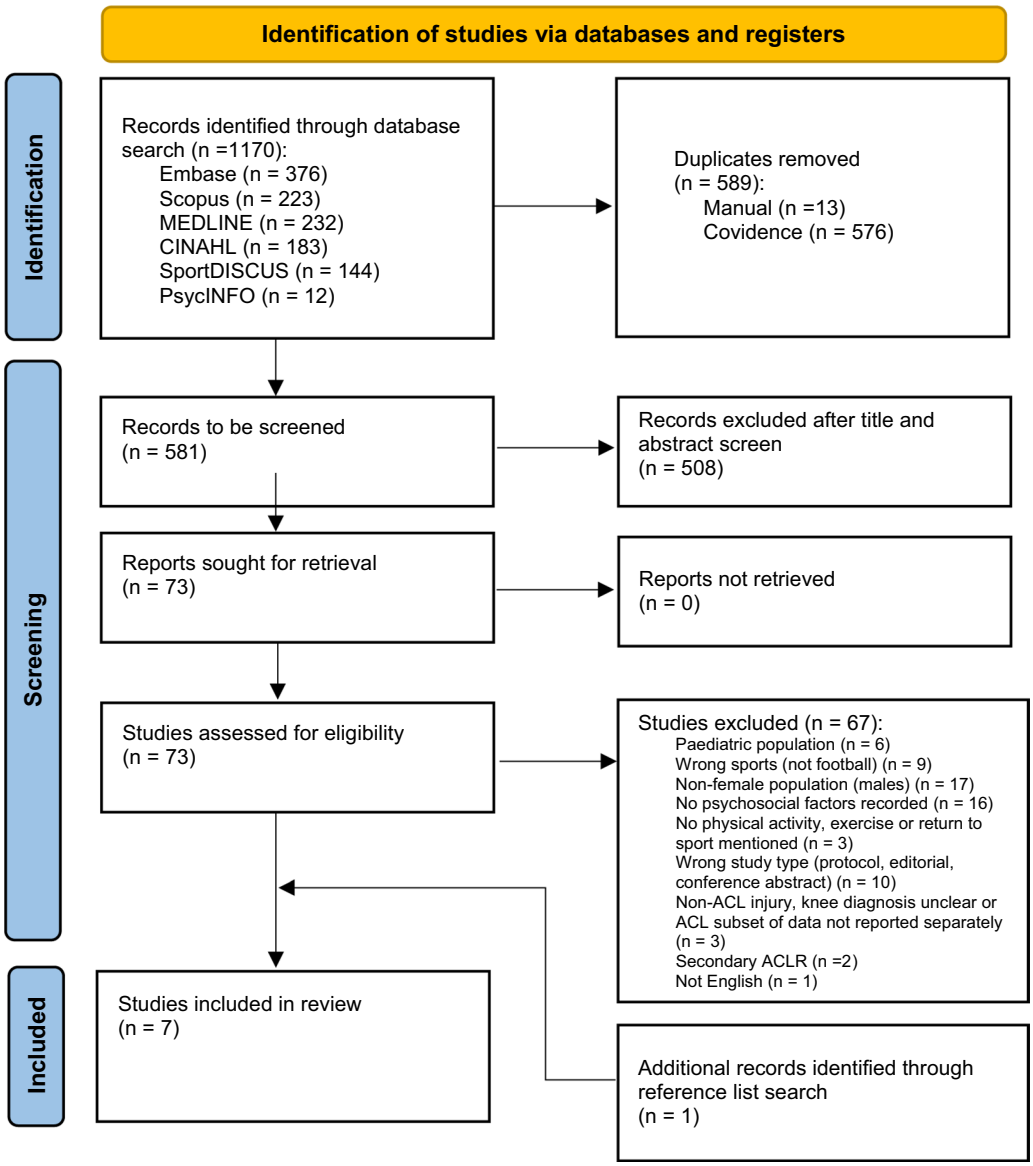


Fig. 1 PRISMA flow diagram of the search and selection process for the scoping review

the female football participant numbers were small (approximately 10%).

Impact of Psychosocial Factors on Return to Sport and Physical Activity

As most studies included in this review were qualitative in design, the psychosocial factors identified should be interpreted as associated with, rather than causal determinants of physical activity and RTS. The included studies identified both psychosocial barriers that hindered RTS and/or physical activity, and facilitators that supported readiness for and subsequent RTS among female footballers following ACL injury. Identified psychosocial factor themes included uncertainty [42, 44], confidence [38–43], fear [38–44], motivation [39, 40, 43], social support [40–42, 44], and athletic identity [40, 41, 43, 44]. While these factors are reported separately and are distinct from each other, they also intersect and overlap. For example, uncertainty coincided with a lack of confidence [43] and can potentially be an outcome of fear [42], whereas when participants were more certain on what to do, they became more confident [40]. Yet despite athletes reporting more certainty when they were physically ready, fear remained a deterrent as participants reported unwillingness to take risks associated with returning to football [43]. However, factors such as motivation, social

Table 3 Summary of study characteristics investigating psychosocial factors on RTS following ACL ligament injury in females

Author (year)	Female athlete number (*n*)	Study aim	Study design & method	Population:		
				Age at time of study (mean)	Football code, level of sport,	Surgical or nonsurgical ACL injury management
Fjellman-Wiklund et al. (2022)	8	To explore the retrospective experiences of suffering an ACL injury, treatment decisions, the individual rehabilitation process and the personal consequences and implications after 20 years or more post-injury	Qualitative, Semi-structured interviews	44 years*	Soccer, level of sport not reported	ACL surgical or non-surgical management
Mahood et al. (2020)	2	To investigate strategies and factors which competitive athletes have used to manage their fear of reinjury after ACLR, in order to resume involvement in competitive sport	Qualitative, Individual interviews in person or online. Web based questionnaires including ACL-RSI and TAS	24.5 years*	Soccer and rugby union, level not reported	ACLR
Kunnen et al. (2020)	9	To understand how athletes defined psychological readiness and knew when they were ready to return to soccer following ACL reconstruction surgery	Qualitative, Open-ended online survey	24 years*	Soccer, local to international level	ACLR
Walker et al. (2022)	2	To investigate the patient-reported barriers and facilitators of ACL reconstruction rehabilitation and identify targets to facilitate participation and adherence to rehabilitation	Qualitative, Focus groups using semi-structured interview guide	37 years*	Local soccer and national touch football	ACLR
Scott et al. (2018)	2	To explore the perspectives of individuals who had undergone ACL reconstruction less than 3 years, to gain a deeper understanding of the impact this injury, surgery and rehabilitation have had on the individual's life, and to describe the factors which influenced the perceived quality of life since the surgery	Qualitative, Semi-structured interviews. Return to sport measured by TAS	34.5 years*	Soccer, level not reported	ACLR

Table 3 (continued)

Author (year)	Female athlete number (*n*)	Study aim	Study design & method	Population:		
				Age at time of study (mean)	Football code, level of sport,	Surgical or nonsurgical ACL injury management
Fältström et al. (2024)	198	To investigate potential sex differences in player-related factors, ACL injury characteristics, and patient-reported outcomes covering knee function, satisfaction with knee function and activity level, and readiness to return to sport in male and female football players 1–3 years after unilateral primary ACLR	Cross-sectional, Data collected preseason (January-April) 2013–2015. Web based questionnaires including ACL-RSI, reason for not RTS, RTS after ACLR, TAS (graded by author off patient reported activity), satisfaction with current activity levels	20.4 years	Soccer; elite, sub elite, recreational level	ACLR
Fältström et al. (2016)	182	To investigate female football players who underwent unilateral primary ACLR to determine whether player-related factors and ACL injury characteristics were associated with the return to playing football. The present study also compared current knee function, readiness to return to sport and knee-related quality of life between those who returned to football and were currently playing and those who had not returned	Cross-sectional, Data collected preseason (January-April) 2013–2014. Web based questionnaires including ACL-RSI, reason for not RTS, motivation scale, RTS after ACLR, TAS (graded by author off patient reported activity), satisfaction with current activity levels	Current players 20.1 years/did not return 20.8 years	Soccer; elite, sub elite, recreational level	ACLR

ACL: anterior cruciate ligament; ACLR: anterior cruciate ligament reconstruction; RTS: return to sport; ACL-RSI: Anterior Cruciate Ligament Return to Sport After Injury scale; TAS: Tegner Activity Score

* Calculated by author (SM) from participant demographic information

support, and athletic identity appeared to act as facilitators to help participants navigate fear related barriers [40, 43].

Uncertainty

We understand participant reference to a lack of knowledge about the next steps in their RTS or progression through rehabilitation, as uncertainty. Uncertainty was identified as a perceived barrier across two qualitative studies [42, 44] and contributed to female footballers returning to a lower competition level than preinjury participation or discontinuing soccer altogether and changing sport entirely. Uncertainty often centred on how to return to physical activity or football [42] and as they progressed through rehabilitation, on how to stay active [44]. Participants reported difficulty seeking external support, finding desirable and acceptable physical activities, and obtaining answers to their questions [42, 44]. Uncertainty is demonstrated in the faltering language of this participant: "It was the physicians who … it was just to give up. It was just… "quit with this [soccer]." It was really upsetting … I was so young so I didn't know any better … this with surgery … if it would have helped so I could have become more physically active …. this is what I think about a lot, if it could have made the situation different as of today?" [42].

Confidence

Confidence was used to describe participants' belief in their own abilities and trust in those supporting their recovery. Conversely, a lack of confidence was characterised by uncertainty, self-doubt, and reduced trust in their ability to return to football. Confidence was identified across three studies, one qualitative [41] and two cross-sectional studies [38, 39], as a perceived barrier for female footballers to RTS. Those who had a lack of trust or confidence in their knee returned to football at a lower competition level or did not return to football at all [38, 39, 41]. Lack of confidence in one's knee is demonstrated in the faltering language of participant quotes reported in some studies: "And I don't think I will ever quite have the confidence in my knee to do stuff like anything that involves that- like, you know, I can feel that, that just in there still, that side way movement, so I don't think I'll ever get back to doing something like that [soccer]" [41].

Confidence was also identified as a perceived facilitator to RTS in three qualitative studies [40, 42, 43]. Specifically, female footballers who had confidence in their healthcare professionals returned to preinjury competition level or a lower competition level [40, 42, 43]. The therapeutic relationship with their healthcare professionals helped them to be confident in the rehabilitation process as they progressed through their goals [42, 43]. As they achieved a certain level of physical capacity, they gained confidence in their physical ability and themselves, and trust in their knee [42, 43]. This is demonstrated by the quote: "My knee was pain free. I could run as far and fast as I could pre-op. I was able to train fully with contact and cut and turn movements for 3 months before I felt confident and comfortable being able to play. Most importantly, I didn't have any knee awareness, and I wasn't scared that the movement that caused my injury would cause me to reinjure it" [43].

Fear

Fear was used to describe concerns about reinjury, sustaining a new injury, or experiencing pain when returning to football or physical activity. It was identified across five studies, three qualitative [41, 42, 44] and two cross-sectional studies [38, 39], as a perceived barrier and reported to have contributed to female footballers altering, limiting, or avoiding physical activity and/or football. Although some participants knew their knee was stable, they could not overcome the fear [44]. As one participant explained: "I mean, my goodness, it's over 20 years ago, and still, whatever you do it's in the back of your head, is it a risk or? What if I get injured again, what if I don't get well? So, you choose what you do" [44].

Female footballers who accepted reinjury risks were more likely to return to football at the same competitive level before injury or a lower competition level, as articulated in three qualitative studies [40, 42, 43]. Participants expressed readiness to return to soccer when they had little self-doubt or fear of reinjury, and confidence in themselves [43]. They linked their ability to overcome fears to consistent, repetitive, and controlled exposure to situations associated with their injury [42]. This desire for competition, commitment towards themselves, their chosen sport, and teammates facilitated RTS despite reinjury risks: "I'm just not that worried" [40]; "[t]he desire to return was stronger than the fear of reinjury" [43]; "[c]onvinc[ed] myself it was worth reinjuring my knee again for the sport I love" [43].

Motivation

Motivation referred to participants' commitment to, enjoyment of, and passion for football, as well as their desire to return to participation following injury. Motivation was identified across two qualitative studies [40, 43] and one cross-sectional study [39] as a perceived facilitator. Female footballers who had strong commitment and/or desire were most likely to RTS at their preinjury level or at a lower competition level [39, 40, 43]. Key motivations included their "love of the game" [43], commitment to sport and their desire to return to the field [40, 43], shown with this quote: "My reason for

returning to sport after that was (…) I have got games left in me, I am not done here, and I have grown to love this sport. I don't really want to be standing on the side lines I would much rather be playing." [40]. They also found motivation in the connection with others, helping them to prepare for RTS [40]. One cross-sectional study articulated players who had returned to soccer had higher motivation, as measured with three questions related to physical activity past and present, to return to football ($p < 0.001$) than those who did not return [39].

Social Support

Social support was described by female footballers in the included studies as either a lack of support, such as feeling like an outsider, or a presence of support, such as having positive support networks around them. Social support was identified as a perceived barrier across three qualitative studies [41, 42, 44], which female footballers appeared to link to their return to football at the same or lower competition level or, for some, discontinuing football altogether. It was perceived to negatively impact female footballers who reported a lack of support or partnership from healthcare professionals during the rehabilitation process, which resulted in them quitting sport or participating at a lower competition level [41, 42, 44]. As participants shared: "Every now and then I think man it would be nice to be checked back up on or something to make sure everything is functioning the way it's supposed to and stuff." [41], "I've given up on physio's a lot of the time…therefore nobody can help me" [41]. Some athletes identified difficulties adjusting with feelings their sporting career had ended [43, 44], and that they were "outsiders" unable to participate with the team [43, 44] and maintain peer relationships [44], leading to social isolation and disconnection.

Across two qualitative studies [40, 41], social support appeared to aid participants. Female footballers who had positive social support networks, including health professionals, family, friends, teammates, and coaches, facilitating their RTS were more likely to RTS at preinjury competition level as defined by TAS [40], although some did return at a lower competition level [41]. Perceived social support and connection reminded participants they were still valued within the team [40]. One participant explained: "And then also the team, they were there to support you whether they knew that I might have mmm, a few nerves around contact situations and they were more than happy to work at a lower level to support me in training and also when it came to game day they are just as excited for you as you are." [40].

Athletic Identity

Athletic identity referred to the degree to which participants viewed football as central to their sense of self, with injury influencing how they perceived themselves as athletes. Athletic identity was identified across two qualitative studies [41, 44], and contributed to female footballers returning to football at their preinjury competition level or lower, as well as modification or avoidance of physical activity and/or football. Female footballers experiencing a loss of identity and independence, RTS of interest at a lower competition level, adjusting to altered physical activity [41] or ending sport career [44]. In some instances, social factors appear to mediate athletic identity: "It was the circle of friends you had when you were training… My friends were in the soccer team so it was really sad when it didn't work out any longer. You had to find other interests… You still had your friends, but they were out in the evenings, at practice." [44].

Female footballers with a strong athletic identity reported in two qualitative studies [40, 43] returned to football at preinjury competition level as defined by TAS [40], although some did return to a lower level than before injury [43]. They articulated a strong desire to play and saw themselves as athletic, competitive, active people [40, 43], acting to facilitate their RTS: "I've always played sports and I'm particularly goal driven; so (…) it was probably completely more of an individual set goal, like I need to get back to playing football" [40]. Alternatively, some of the participants who did not RTS found new roles within the sport such as coaching, finding an alternative to maintain interests [44].

Discussion

The aim of this scoping review was to investigate the role of psychosocial factors in females participating in any football code post ACL injury, and the perceived impact on return to physical activity and sport. Six psychosocial factors were identified during the evidence synthesis: uncertainty, confidence, fear, motivation, social support, and athletic identity. To the best of our knowledge, this is the first scoping review which has examined psychosocial factors in female football players specifically related to ACL injury and RTS or physical activity. Results highlighted the importance of psychosocial factors in RTS and physical activity in female athletes post-ACL injury, indicating that a more holistic approach to exercise prescription, rehabilitation strategies and RTS decisions for female footballers is needed. The review findings can be used by sport medicine professionals, coaches, and sport trainers to inform support and resources for female footballers returning to physical activity and/or RTS.

Key findings from this review were that psychosocial factors can act not only as barriers in RTS and physical activity but can facilitate female footballers RTS or physical activity after ACL injury. Fear, uncertainty, lack of confidence, and limited social support were perceived as barriers to RTS, leading to lower levels of sport participation and prompting individuals to modify, limit, or avoid physical activity and football altogether. In contrast, growing confidence, motivation, strong social support, and athletic identity contributed to female football players RTS at their preinjury level and at a lower competition level.

In this scoping review, fear of pain, and reinjury emerged as a predominant barrier to both RTS and ongoing physical activity in female footballers. Fear of reinjury illustrates how psychological factors can be shaped by social influences. For example, pressure from coaches and teammates to return and perform was associated with anxiety, reduced focus and concentration, and heightened concerns about sustaining another ACL injury [46]. Consistent with the BPS Model of Athletic Rehabilitation, social pressures and psychological responses may contribute to intermediate biopsychological outcomes that ultimately influence sport injury rehabilitation outcomes [17]. For example, female footballers frequently reported fear and anxiety despite satisfactory physical recovery, suggesting a disconnect between physical recovery and readiness to RTS [46]. Previous research has consistently reported fear as a key psychosocial barrier for returning to sport in mixed-sex athletes following traumatic knee injury, including after ACLR [15, 22]. Fear can impede RTS and physical activity through several pathways, such as pain [47], fear of the injury itself [48], catastrophising [49], and reinjury anxiety (i.e. negative thoughts about the consequences of injury) [50]. In male and female athletes recovering from an ACL injury, cutting has been reported as a common fear-evoking task in competitive sport [51]. As a dominant movement required to participate in the football codes and a common mechanism of ACL injury [52], this may drive avoidance of essential sport tasks and suppress sport participation in female football players. In contrast, when female footballers in this current review accepted the reinjury risks, they were more likely to RTS. Measuring and including fear of pain and/or reinjury as a rehabilitation target alongside physical knee outcomes, particularly when initiating the RTS phase, is recommended [48] to support female footballers recovering from an ACL injury.

Fear of reinjury is related to an athlete's confidence [50], which was found in this review as both a barrier and facilitator for female footballers in RTS and physical activity. A lack of confidence in themselves, their health professionals, and trust in the knee was identified across six of the seven studies in this scoping review and was associated with RTS at lower level of competition or not returning at all. Previous research indicates confident athletes at the time of RTS after ACLR were more likely to meet RTS criteria, such as higher physical function, strength, and performance [53]. Specifically, greater quadricep strength was associated with a greater psychological readiness and therefore confidence in RTS in football players with a noncontact injury mechanism [54], highlighting the reciprocal relationship between psychological factors and intermediate biopsychological outcomes within the BPS Model of Athletic Injury Rehabilitation [17]. In this review, uncertainty appeared to be related to female footballer players' lack of confidence, particularly regarding how to navigate the RTS and physical activity process, leading to returning to a lower level of football or a complete change in sport. In contrast, athletes who expressed higher confidence in their healthcare professionals reported gaining trust in their knee through objective milestones of physical capacity. Rehabilitation protocols following ACLR are well established, although athletes from both sexes still feel uncertain about RTS [55]. Previous research similarly shows that uncertainty permeates life after ACLR for both males and females, related to treatment, physical capacity, future sport participation, and long-term consequences and/or limitations [56]. Addressing fear of reinjury may require strategies beyond physical rehabilitation alone. Female footballers identified teammates working at a lower level of technical capacity to expose them to contact situations helped prepare them for RTS [40]. Psychologically informed practice, for example, uses exposure based approaches to target fear-related impairments and inform the design of appropriate graded exposure interventions [51]. Considering physical milestones alongside psychosocial factors such as confidence, uncertainty, and fear, and how the interactions impact RTS in female football players may support these athletes.

Psychosocial factors like motivation, social support, and athletic identity helped female footballers navigate fear-related barriers, thereby facilitating RTS and physical activity. Our review found female football players who had positive social support and team connection were more likely to RTS at preinjury competition level and reported feeling valued within their team environment, which in turn further mediated their athletic identity and helped mitigate fear. Consistent with this, regardless of sex, greater connectedness, and sense of belonging were associated with a positive perception on RTS [57]. Moreover, both social support and athletic identity have been shown to have an impact on rehabilitation engagement and RTS outcomes in male and female athletes,

including football players [58]. A strong sense of belonging, facilitated by social connection and athletic identity plays a crucial role in rehabilitation and RTS after ACL injury. While a strong athletic identity can support motivation and social support, it may represent a double edged sword, offering short term motivational benefits while posing longer term psychological risks if not balanced with a broader sense of self [59]. Without establishing a unique personal identity through varied experiences and interests, athletes may struggle to adjust to cessation of sport and coping poorly with periods of injury and rehabilitation [60]. This may also lead to tendencies to fast-track rehabilitation and disregard treating practitioner recommendations [59]. However, a stronger athletic identity may also improve coping and higher levels of physical function in some athletes [59]. Support from teammates, coaches, and health professionals helps athletes stay engaged in their sporting environment, reinforcing their value within the team and preserving a sense of identity and purpose [61–63]. This sense of belonging enhances psychological wellbeing, motivation, and adherence to rehabilitation [61, 62]. More broadly, belonging promotes positive social relationships and emotional regulation, while its absence increases risks of poor mental and physical health, and social isolation [26, 33]. For female footballers, these dynamics suggest that social support and athletic identity foster belonging, which may be a key mechanism enabling successful return to sport after ACL injury. Particularly for female athletes, external support systems, such as engaging with the team, have been reported as crucial after ACLR [56], which contrasted with males who reported experiencing isolation from social and team engagements as a consequence of physical limitations during recovery [64]. This sex difference may be due to limited opportunities to partake in social activities during ACLR rehabilitation and a stronger internal locus of control [60] and self-efficacy [16, 20, 56]. Supporting psychological needs and social support positively impacts participation and strengthening commitment to sport [65]. Considering these differing demands related to social support and maintaining athletic identity within rehabilitation and RTS planning may improve psychological wellbeing and better support female football players in returning to sport. More broadly, these findings support the BPS Model of Athletic Injury Rehabilitation by highlighting how interactions between psychological and social factors influence rehabilitation experiences and RTS outcomes beyond physical recovery alone.

The findings from this scoping review identify more research is needed to investigate the female experience in sport and how psychosocial factors influence RTS outcomes for female footballers. Only seven studies met inclusion criteria across all football codes, revealing a notable gap in the evidence base. Many methodologically sound studies that included female participants were excluded because study findings were not reported separately by sex; notably, 17 studies (25%) did not report separate results for female athletes. The findings in this review were identified through mostly mixed sex studies (86%) using qualitative methods such as focus groups, semi-structured interviews, or open-ended surveys that were not tailored to the experiences of female football players. This reflects a broader pattern in sport and exercise science research, where only 6% of studies focus exclusively on females and 35% include females with comparison against to male norms [66]. Future mixed sex studies should therefore distinguish between male and female subset data to avoid potential gendered assumptions and to capture the unique articulation of injury experiences, needs, contexts and meanings in females. This is critical given evidence that gender has been considered a predictor of fear/anxiety of reinjury, with women reporting higher levels of fear/anxiety than men [50]. While both males and females experience fear following ACLR, females exhibited greater fear with general physical activity, whereas males expressed more fear related to sport-specific movements [67]. Similar gendered patterns have been noted in ACL injury and concussion research, where female athletes face environments that shape recovery experiences differently from their male counterparts [66]. A recent systematic review highlighted the impact of psychosocial factors in mixed sex athletes with concussion, finding factors similar to this scoping review such as fear, social support and identity along with perceived emotional factors were important determinants in RTS and level of sport, yet clinicians receive little guidance on how to address these factors [68].

Investigating RTS following an ACL injury in female football players is important given their lower RTS rates and higher reported levels of prolonged physical and psychosocial impacts compared to males. Females have a 25% lower rate of RTS within the first five years after ACLR compared with males, along with poorer knee-related function and greater psychological distress [69]. A systematic review by Figueroa et al. [70] reported a wide range of RTS success among female soccer players (50–90%) and high proportion of preinjury RTS levels (75–100%). However, while 90% of national level soccer players returned to sport, many played fewer games and minutes post-injury [71]. The reasons for failure to RTS remain unclear as only four of the 15 included studies in the systematic review reported relevant data and often described reasons vaguely, such as 'not knee related' reasons or psychological reasons such a fear

of reinjury or lack of knee confidence [70]. This inconsistency suggests the need for more standardised and comprehensive reporting on RTS barriers in female athletes. Further, scheduling pressures in football have been substantiated in the literature, with injured female players competing in almost twice as many back-to-back fixtures [72]. In contrast to male footballers, female footballers often face short term contracts and financial pressures, as well as extensive travel, reduced recovery windows and insufficient load monitoring [73]. These contextual demands highlight that rehabilitation following ACL injury occurs within a broader sociocultural environment. The pressures associated with short-term contracts, financial insecurity, and demanding competition schedules may influence psychological wellbeing, perceptions of support, rehabilitation engagement, and RTS decision-making in female footballers. This is further compounded by the gendered research environment in which female footballers remain underrepresented, limiting the applicability of literature to this population [30] and perpetuating a reliance on findings derived predominantly from male dominated research [66].

Ensuring female athletes return to full physical activity and RTS is a key consideration after ACLR or injury [70, 74] given the significant long-term health implications. Female athletes with previous ACL injury have four to six times greater risk of knee osteoarthritis (OA) and over twice the risk of needing a total joint replacement within 15 years compared to female athletes without ACL injury [75]. Knee OA is linked with reduced physical activity, greater sedentary behaviours and increased risk of becoming overweight 3–10 years after ACLR [76]. Participation in physical activity has been identified as a modifiable risk factor in preventing chronic diseases and supporting psychological wellbeing [14, 77]. Although the benefits of physical activity and reducing sedentary behaviour largely outweigh the associated risks, athletes with chronic conditions like knee OA after ACL injury experience fear of symptom exacerbation and worsening their condition, reduced motivation and diminished confidence [77]. A recent consensus statement on risks of physical activity for people living with long-term conditions advocated for continued engagement in appropriately prescribed physical activity, emphasising its overall benefits [77]. The present review suggests that psychosocial factors including fear, motivation and confidence influence participation in physical activity and RTS long before the presence of chronic conditions like knee OA emerge. Addressing psychosocial factors during ACL injury rehabilitation may facilitate more female football players RTS and physical activity, minimising long term risks and chronic health conditions.

Haberfield et al. [69] utilised a mixed-methods concept mapping approach to propose a seven-cluster framework derived from the perspectives of both patients and practitioners to address the gender disparity in outcomes following an ACL injury in women. This framework prioritises fostering goal-driven rehabilitation and promoting mental and emotional wellbeing, while also addressing the need for adaptable environments, targeted education, and multidisciplinary team engagement [69]. While clusters related to peer, group, and social support were rated lower in relative importance and feasibility, they remained fundamental to sustaining motivation and adherence to rehabilitation and recovery [69]. This current scoping review supports this framework, identifying fear, motivation, uncertainty, social support, the disruption of athletic identity and the need to bolster confidence as key psychosocial factors shaping the RTS process after ACL injury among female footballers. Together, both studies identify the need for future research to explore unique female psychosocial factors and develop sex-specific approaches and clinical guidelines to better support healthcare practitioners and female athletes to promote successful RTS and physical activity participation following ACL injury.

Clinicians treating female footballers after ACL injury should consider incorporating routine screening for psychosocial factors alongside rehabilitation and RTS decision making. Given the perceived impact of factors such as confidence, motivation, fear and social support on recovery experiences identified in this review, regular assessment of psychosocial wellbeing may help clinicians identify barriers and facilitators to recovery and provide targeted support throughout rehabilitation and RTS planning. Referral to a sport psychologist or mental health professional may be warranted when psychosocial risk factors are identified and may have a positive effect on female athletes returning to preinjury level of sport [78]. Further research is needed to evaluate screening tools, referral pathways, and intervention strategies to determine the most effective ways to identify and address psychosocial factors impacting RTS in female footballers after ACL injury. Future research should also consider gendered sporting environments and potential sex differences in psychosocial factors through a focus on sex and gender in study methodology and reporting practices. Such work is critical to advancing understanding of the impact of psychosocial factors on RTS outcomes and ongoing participation in physical activity in female footballers after ACL injury.

Strengths and Limitations

This scoping review had several strengths. It was the first, to our knowledge, to investigate psychosocial factors impacting return to physical activity and sport in female football players following an ACL injury. The review was performed according to the PRISMA-ScR checklist, which added to the methodological rigour of our review, following a prospective protocol publication, and followed a comprehensive systematic search strategy. The review also contained both qualitative and quantitative data from varying age groups and time since ACL injury, which increased the generalisability of findings. In addition, this review identified areas for future research to further support females in RTS and physical activity. Despite these strengths, this scoping review does have limitations. The review did not differentiate gender in questions, data collection, methodologies or analytic processes, which may have introduced bias into the results. Given the broad age range of included participants in this study, age-related heterogeneity should be considered when interpreting findings. Only English-language articles were included. In addition, as scoping reviews do not evaluate methodological quality, the included studies were not critically analysed for quality which may have created an information bias. A further limitation of this review is that given the predominantly qualitative nature of the included studies, we were unable to determine causal relationships between psychosocial factors and RTS or physical activity outcomes. In addition, inconsistent measurement of RTS and physical activity limited the ability to draw direct conclusions regarding the association between perceived psychosocial factors and these outcomes. Furthermore, potential differences in recovery experiences between participants with and without ACLR, and those who underwent concomitant procedures such as meniscal repair, were not specifically explored. Future research employing longitudinal and quantitative designs is needed to examine clinical subgroup differences, clarify the relationships between psychosocial factors and RTS and physical activity outcomes, and establish the direction and causality of these associations.

Conclusion

This scoping review highlights the importance of psychosocial factors in shaping RTS and physical activity experiences among female football players following ACL injury. Fear, uncertainty, lack of confidence and poor social support were identified as perceived barriers, whereas confidence, motivation, strong social support and athletic identity were identified as perceived facilitators. These findings support consideration of psychosocial factors alongside physical recovery during rehabilitation and RTS decision-making. Further high-quality, female-focussed research is needed to better understand these relationships and inform targeted screening and intervention strategies, with greater consideration of sex and gender in study design, methodology, and reporting.

Abbreviations

ACL	Anterior cruciate ligament
ACLR	Anterior cruciate ligament reconstruction
RTS	Return to sport
PRISMA-ScR	Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for scoping reviews
PCC	Population Concept Context
ACL-RSI	Anterior Cruciate Ligament Return to Sport after Injury scale
TAS	Tegner Activity Score
OA	Osteoarthritis

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Author Contributions

SM made contributions to the conceptualisation and study design, screening, data extraction, writing and drafting intellectual content. ED made contributions to design of the study, screening and critical feedback. JF made contributions to the design of the study, cross checking data extraction and interpretation of data and critical feedback. JH made contributions through critical feedback of the manuscript. SM, ED and JF agreed to be accountable for all aspects of the work. All authors (SM, ED, JF, and JH) have read and approved the final version of this manuscript.

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Data Availability

The dataset supporting the conclusions of this article are included within the article and its additional files.

Declarations

Ethics Approval and Consent to Participate

Not Applicable.

Consent for Publication

Not applicable.

Competing Interests

Skye Meredith, Jane Hunt, Jessica Farley and Elsa Dent declare no competing interests.

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